

**AI in Government: IRS Fraud Detection Case Study**

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## **Introduction**

Artificial intelligence is becoming an increasingly important tool in government because public agencies must manage large volumes of information while operating with limited time, staff, and financial resources. In the United States, the Internal Revenue Service (IRS) is responsible for administering federal tax laws, processing tax returns, collecting revenue, issuing refunds, and promoting taxpayer compliance. Because the agency receives an enormous volume of financial information each year, it is not possible for human auditors to manually examine every return in detail. This creates a need for technologies that can identify suspicious patterns, prioritize cases, and help public employees use their expertise more efficiently.

The IRS uses artificial intelligence, machine learning, and advanced data analytics to support tax compliance and fraud detection. These systems can compare tax returns with information from employers, financial institutions, and other third-party sources; identify unusual deductions or reporting inconsistencies; and estimate which cases present the highest risk of noncompliance. AI has also been used to support the selection of complex partnership and high-income cases for examination and to detect patterns associated with fraudulent refund claims. Rather than independently determining guilt or imposing penalties, these tools assist IRS employees by directing attention toward returns that may require further human review.

The adoption of AI in tax administration offers important public benefits, including more efficient audit selection, improved detection of fraudulent activity, and better allocation of government resources. However, it also raises concerns about false positives, taxpayer privacy, algorithmic bias, limited transparency, and the consequences of allowing automated risk scores to influence government investigations. This report examines the IRS as a real-world case study of AI in the public sector by analyzing the problem addressed, the technologies used, the benefits

achieved, and the limitations of the current approach. It then proposes an Explainable AI Review System that would provide authorized auditors with clear risk factors, confidence scores, documented audit trails, and structured human review before significant enforcement actions are taken. The central argument is that AI can strengthen tax administration, but its legitimacy depends on combining analytical efficiency with transparency, accountability, and meaningful human oversight.

### **Case Study Analysis**

The Internal Revenue Service (IRS) is the federal agency responsible for administering and enforcing the United States tax system. Each year, the agency processes hundreds of millions of tax returns submitted by individual taxpayers, businesses, and organizations. Given the enormous volume of financial information received annually, manually reviewing every return would be impossible. At the same time, tax fraud, underreported income, identity theft, and fraudulent refund claims cost the federal government billions of dollars each year. These challenges have encouraged the IRS to adopt artificial intelligence, machine learning, and advanced data analytics to improve tax compliance while making more efficient use of limited auditing resources.

Rather than replacing human auditors, the IRS uses AI as a decision-support tool that helps identify tax returns with a higher probability of noncompliance. Machine learning models analyze large amounts of historical tax data alongside information received from third-party sources, including employers, financial institutions, and other government agencies. The system searches for patterns, anomalies, and inconsistencies that would be extremely difficult for human reviewers to detect manually. Examples include income that does not match third-party records, unusually high deductions compared with similar taxpayers, repeated business losses in profitable industries,

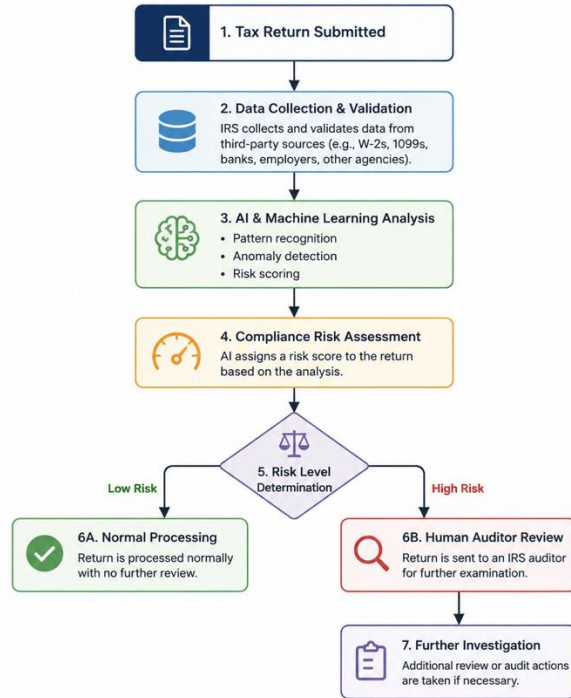
suspicious cryptocurrency transactions, and offshore financial activities that may require additional investigation.

One of the primary functions of AI within the IRS is risk scoring. Instead of determining whether a taxpayer has committed fraud, the system estimates the likelihood that a particular tax return deserves further examination. Returns with higher risk scores receive greater attention from auditors, allowing the agency to prioritize investigations where potential noncompliance is most likely. This approach enables the IRS to allocate its personnel more effectively while reducing the need to randomly select returns for audit.

Artificial intelligence also plays an important role in detecting refund fraud before payments are issued. Machine learning systems evaluate returns shortly after they are submitted, identifying characteristics that resemble previously known fraud schemes. If a return is flagged as suspicious, the refund may be temporarily delayed while additional verification is performed. This process helps prevent fraudulent refunds from being distributed while protecting public funds from financial losses.

Beyond fraud detection, AI supports criminal investigations by identifying patterns that may indicate intentional tax evasion rather than simple reporting mistakes. The system assists investigators in distinguishing routine civil compliance issues from cases that may require criminal enforcement. This capability allows investigators to focus their efforts on the most serious violations while improving the overall efficiency of tax administration. Figure 1 summarizes the general workflow of the IRS's AI-assisted tax return review process.

Figure 1. Current IRS AI-Assisted Tax Return Review Process



Source: Created by the author based on IRS and GAO documentation.

The adoption of AI has produced several important benefits for the IRS. First, it significantly improves operational efficiency by allowing millions of tax returns to be analyzed within a relatively short period of time. Human auditors can then concentrate on the highest-risk cases instead of manually reviewing large numbers of compliant returns. Second, AI improves the agency's ability to detect sophisticated fraud schemes by recognizing complex relationships across multiple datasets that would be difficult for humans to identify independently. Finally, better audit selection contributes to more effective use of public resources by directing enforcement efforts toward cases with the greatest potential financial impact.

Despite these advantages, the current approach also presents important limitations. Machine learning models are not perfect and may generate false positives, causing honest taxpayers to undergo unnecessary audits or experience delays in receiving legitimate refunds.

Conversely, false negatives may allow fraudulent returns to go undetected if they do not match previously identified patterns. The accuracy of these systems depends heavily on the quality, diversity, and continuous updating of the training data used to develop the models.

Another significant challenge involves transparency. Many AI models operate as complex predictive systems whose internal reasoning is difficult for both auditors and taxpayers to fully understand. If a taxpayer is selected for additional review based largely on an AI-generated risk score, questions naturally arise regarding how that decision was made, which factors influenced the assessment, and whether similar taxpayers are being evaluated consistently. Maintaining public trust therefore requires careful human oversight and clear governance policies regarding the role of AI in tax administration.

Finally, privacy remains an important consideration because the IRS analyzes highly sensitive financial information from millions of individuals and businesses. Although the agency already operates under strict legal and cybersecurity requirements, the growing use of AI increases the importance of responsible data governance, secure information management, and appropriate safeguards against unauthorized access or misuse. As AI continues to become more sophisticated, balancing fraud prevention with taxpayer rights will remain one of the most important challenges facing modern tax administration.

### **Innovative Proposal**

While the IRS has significantly improved tax compliance through artificial intelligence and advanced data analytics, its current approach could be strengthened by increasing transparency and supporting more consistent decision-making during the audit selection process. I propose implementing an Explainable AI Review System, an additional decision-support layer designed to complement the existing fraud detection models rather than replace them. The goal of this system

would be to improve accountability, strengthen public trust, and provide auditors with clearer insights into why a tax return has been identified as potentially noncompliant.

Under this proposal, AI would continue performing its current tasks of analyzing tax returns, identifying unusual patterns, and generating compliance risk scores. However, instead of presenting auditors with only a numerical risk assessment, the system would also generate an explanation highlighting the primary factors that contributed to the recommendation. For example, the platform could identify inconsistencies between reported income and third-party records, deductions that significantly exceed industry averages, repeated business losses, or unusual financial transactions that differ from historical taxpayer behavior. These explanations would allow auditors to understand the reasoning behind the AI's assessment before making any enforcement decisions.

Another important feature would be the inclusion of confidence scores alongside each recommendation. Rather than treating every flagged return equally, auditors could evaluate both the estimated compliance risk and the model's confidence in its prediction. Cases with lower confidence could receive additional review before further action is taken, helping reduce unnecessary audits while allowing investigators to focus on cases supported by stronger evidence. This approach encourages more informed decision-making instead of relying solely on automated outputs.

The proposed system would also strengthen accountability by creating a secure audit trail documenting each stage of the review process. The system would record the factors identified by the AI, the confidence level assigned to the case, the actions taken by the auditor, and the final outcome of the investigation. Maintaining these records would support internal quality assurance,

facilitate future model improvements, and provide greater transparency for oversight bodies responsible for evaluating government AI systems.

Human oversight would remain a central component of the proposal. The AI would never make final enforcement decisions or automatically determine taxpayer guilt. Instead, every significant action—including audit selection, refund delays, or criminal referrals—would continue requiring review and approval by qualified IRS personnel. Preserving human judgment helps ensure that AI functions as an advisory tool while reducing the risks associated with false positives, algorithmic bias, and overreliance on automated recommendations.

Although implementing an Explainable AI Review System would require additional investment, technical development, and employee training, the long-term benefits could outweigh these challenges. Providing clearer explanations, confidence scores, and documented decision processes would improve the consistency of investigations while reinforcing taxpayer confidence in the fairness of AI-assisted enforcement. As governments continue adopting artificial intelligence, combining technological efficiency with transparency and human oversight will be essential for building trustworthy public institutions capable of using AI responsibly.

## **Conclusion**

Artificial intelligence has become an increasingly valuable tool for improving government operations by helping public agencies analyze large volumes of information, allocate resources more efficiently, and identify complex patterns that would be difficult to detect manually. The IRS demonstrates how AI, machine learning, and advanced data analytics can strengthen tax compliance by supporting audit selection, identifying suspicious tax returns, and improving the detection of fraudulent activities. Rather than replacing human expertise, these technologies enable



auditors to focus their efforts on the cases that present the highest compliance risk while improving the overall efficiency of tax administration.

At the same time, the use of AI in government must be carefully managed to ensure fairness, transparency, and accountability. Challenges such as false positives, privacy protection, algorithmic bias, and limited explainability highlight the importance of maintaining meaningful human oversight throughout the decision-making process. Public trust depends not only on the effectiveness of AI systems but also on citizens' confidence that government decisions are made responsibly and can be justified.

The proposed Explainable AI Review System demonstrates one possible path toward more responsible AI adoption within the IRS. By combining explainable recommendations, confidence scores, documented audit trails, and mandatory human review, the system could strengthen both operational efficiency and public confidence. As artificial intelligence continues to evolve, its greatest value in the public sector will not come solely from automation, but from supporting transparent, accountable, and trustworthy government services that better serve both public institutions and the citizens they represent.

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### **Author's Note**

Google NotebookLM and ChatGPT were used as learning and research support tools throughout the development of this assignment. These tools assisted with organizing course materials, exploring concepts, and deepening understanding of topics discussed in class, including AI advancements, ethics, governance, and future trends. They were used to support the learning process rather than as primary sources of information.